Today's agenda

- Intro to 2D Arrays
- Print matrix row wise
- Print matrix Col wise
- Print matrix in wave form
- Array list → {Dynamic arrays}

AlgoPrep

//Intro

↳ Mid semester exam (DSA) : marks

Column

	(100) foundation	(100) level	(100) competitive
0	99	100	98
1	90	95	93
2	99	100	98
⋮			
99	40	30	25

row

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//Syntax

```
int[][] arr = new int [5][4];
```

row col

arr

	0	1	2	3
0	0	0	0	0
1	0	0	0	~~0~~ 30
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0

arr[1][3] = 30;

arr[3][2]

5*4 = 20 elements

Q) Print matrix rowwise

↳ Print the given mat[n][m] rowwise.

arr[4][5]:

	0	1	2	3	4
0	10	20	30	40	50
1	60	70	80	90	100
2	110	120	130	140	150
3	160	170	180	190	200

4x5

10 20 30 40 50
60 70 80 90 100
110 120 130 140 150
160 170 180 190 200

i j

0,0 1,0 2,0 3,0
0,1 (j+1) 1,1 (j+1) 2,1 3,1
0,2 (j+1) 1,2 (j+1) 2,2 3,2
0,3 (j+1) 1,3 (j+1) 2,3 3,3
0,4 (j+1) 1,4 (j+1) 2,4 3,4

```
for (int i=0; i<n; i++){
    for( j=0; j<m; j++){
        System.out.print(arr[i][j] + " ");
    }
    System.out.println();
}
```

T.C: O(n*m)

S.C: O(1)

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```
for (int i=0; i<n ; i++){
    for( j=0; j<m; j++){
        System.out.print(arr[i][j] + " ");
    }
    System.out.println();
}
```

n=4
m=5

arr

	0	1	2	3	4
0	10	20	30	40	50
1	60	70	80	90	100
2	110	120	130	140	150
3	160	170	180	190	200

i → 2

4×5

j

10 20 30 40 50
60 70 80 90 100
110 120 130 140 150
160 170 180 190 200

i: 0
j: 0, 1, 2, 3, 4, 5 exit

i: 1
j: 0, 1, 2, 3, 4, 5 exit

i: 2
j: 0, 1, 2, 3, 4, 5 exit

i: 3
j: 0, 1, 2, 3, 4, 5 exit

4
exit

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```java
public class Main {
    public static void main(String[] args) {
        Scanner scn = new Scanner(System.in);
        // first two integers -> No. of rows, No. of cols
         int n = scn.nextInt();
         int m = scn.nextInt();
         int[][] arr = new int[n][m];

         for(int i=0;i<arr.length;i++){
             for(int j=0;j<arr[0].length;j++){
                 arr[i][j] = scn.nextInt();
             }
         }



         // No of rows -> arr.length == n
         // No of cols -> arr[0].length == m
         for(int i=0;i<arr.length;i++){
             for(int j=0;j<arr[0].length;j++){
                 System.out.print(arr[i][j] +" ");
             }

             System.out.println();
         }
     }
}
```

```java
public class Main {
    public static void main(String[] args) {
        //creating 2D array without Input

        // 1D Array - int[]arr = {10,20,30,40};

        int[][] arr = {{10,20,30,40},
                       {50,60},
                       {90,100,110,120, 130}};

        System.out.println(arr[0].length);
        System.out.println(arr[1].length);
        System.out.println(arr[2].length);

    }
}
```

Break till 9:34 pm

Q) Print matrix colwise

↳ Print the given mat[n][m] colwise.

arr[4][5]:

	0	1	2	3	4
0	10	20	30	40	50
1	60	70	80	90	100
2	110	120	130	140	150
3	160	170	180	190	200

10 60 110 160
20 70 120 170
30 80 130 180
40 90 140 190
50 100 150 200

i j

0 0 0 1 0 2 0 3 0 4
1 0 1 1 1 2 1 3 1 4
2 0 2 1 2 2 2 3 2 4
3 0 3 1 3 2 3 3 3 4

```
for (int j=0; j<m; j++){
    for (int i=0; i<n; i++){
        System.out.print(arr[i][j] + " ");
    }
    System.out.println();
}
```

T.C: O(m*n)

S.C: O(1)

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Q) Print matrix in wave form

↳ Print the given mat[n][m] in wave form.

arr[4][5]:

	0	1	2	3	4	
0	10	20	30	40	50	L-R → 10 20 30 40 50
1	60	70	80	90	100	R-L → 100 90 80 70 60
2	110	120	130	140	150	L-R → 110 120 130 140 150
3	160	170	180	190	200	R-L → 200 190 180 170 160

L-R → 0, 2, 4 ... (even row)

R-L → 1, 3, 5 ... (odd row)

even no: no. divisible by 2

```
for (int i=0; i<n; i++){
    if (i%2==0){
        for( j=0; j<m; j++) {
            System.out.print(arr[i][j] + " ");
        }
    }
    else {
        for( j=m-1; j>=0; j--) {
            System.out.print(arr[i][j] + " ");
        }
    }
    System.out.println();
}
```

T.C: O(m*n)

S.C: O(1)

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	0	1	2	3	4
0	10	20	30	40	50
1	60	70	80	90	100
2	110	120	130	140	150
3	160	170	180	190	200

4x5

```
for (int i=0; i<n ; i++){
    if (i%2==0){
        for( j=0; j<m ; j++){
            System.out.print(arr[i][j] + " ");
        }
    }
    else {
        for( j=m-1; j>=0; j--) {
            System.out.print(arr[i][j] + " ");
        }
    }
    System.out.println();
}
```

i	j
0	0 1 2 3 4 5
	usefit
1	4 3 2 1 0
2	0 1 2 3 4 5

10 20 30 40 50

→ 100 90 80 70 60

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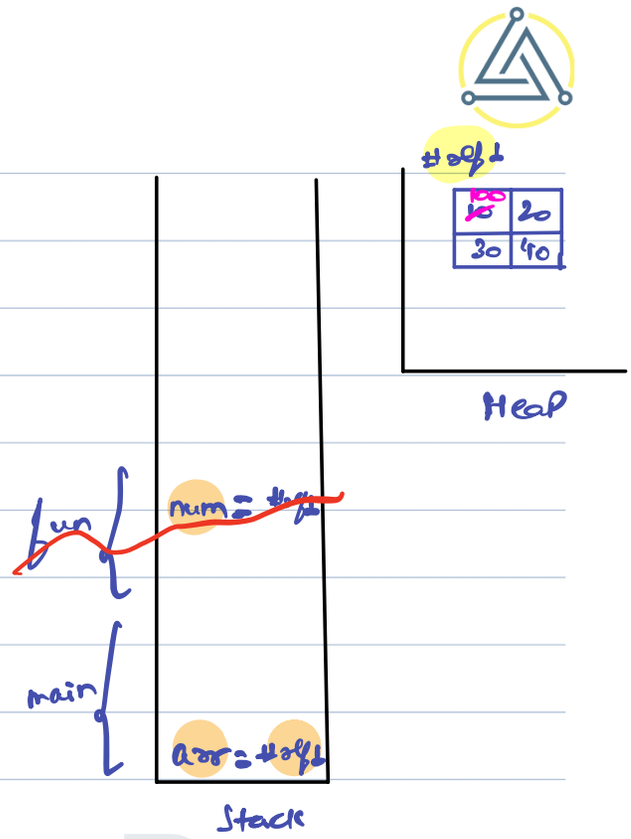
```
void main (){

    int [][] arr = { {10,20} , {30,40} };

        System.out.println(arr[0][0]); -> 10
            fun(arr);
        System.out.println(arr[0][0]); -> 100
}

Public static void fun (int[][] num){

        num[0][0] = 100;
}
```



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int[] arr = new int[10];

// Arraylist or Dynamic array

name

```
ArrayList<Integer> list = new ArrayList<>();
```

0	1	2
10	20	30

→ list.add(10);
→ list.add(20);
→ list.add(30);

→ T.C : O(1)

// To access index

```
S.O.P (list.get(1));
```
→ 20

↳ T.C : O(1)

// Update the value

```
list.get(2) = 200;
list.set(1, 200);
```

↳ T.C : O(1)

// No. of element

```
list.size();
```

↳ T.C : O(1)

// remove index

```
list.remove(idx);
```

remove last idx → O(1)

remove 0th idx → O(N)

list

0	1	~~2~~	~~3~~ 2	~~4~~ 3
10	20	~~30~~	40	50

```
list.remove(2);
```

→ Arraylist in the background is just Array being used smartly.

// Doubt Session

mandiff2: $\max(arr[i] - arr[j] + i - j)$

mandiff3: $\max(arr[i] - arr[j] + j - i)$

mandiff4: $\max(|arr[i] - arr[j]| + i - j)$

$\max(arr[i] - arr[j], -(arr[i] - arr[j]))$

$\Downarrow$

$\max(arr[i] - arr[j], arr[j] - arr[i])$

$\Downarrow$

$|x| = \max(x, -x)$

$|8| = \max(8, -8) = 8$

$|-8| = \max(-8, -(-8)) = 8$

$\max \begin{cases} \cancel{arr[i] - arr[j] + i - j} \\ \cancel{arr[j] - arr[i] + i - j} \end{cases}$

int

int

$i + j$ swap

$arr[i] - arr[j] + j - i$

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$\{10 \quad 20 \quad 13 \quad 3 \quad \overset{d_i}{1} \quad 35\}$

max : $-\infty \quad \cancel{10} \quad 20$

2nd max : $-\infty \quad -\infty \quad \cancel{10} \quad 13$

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